1.9 Compositions of Matrix Transformations

This section discusses the analogs of matrix multiplication and matrix inversion for matrix transformations, and illustrates these ideas with the familiar geometric operations of rotation, reflection, and projection. A key by-product is an explanation of why matrix multiplication is defined in the way it is.

Compositions of Matrix Transformations

The **composition** of two matrix transformations is the process of applying one transformation and then applying another to the result. If $T_A : R^n \to R^k$ and $T_B : R^k \to R^m$, then applying T_A first and T_B second creates a new transformation from R^n to R^m called the **composition of T_B with T_A**, denoted $T_B \circ T_A$:

$$(T_B \circ T_A)(\mathbf{x}) = T_B(T_A(\mathbf{x})) \tag{1}$$

$$\begin{tikzcd} R^n \arrow[r, "T_A"] \arrow[dr, dashed] & R^k \arrow[r, "T_B"] & R^m \\ \mathbf{x} \arrow[r] & T_A(\mathbf{x}) \arrow[r] \arrow[ur, dashed] & T_B(T_A(\mathbf{x})) \\ & T_B \circ T_A & \end{tikzcd}$$

The following theorem is the key result: it shows that the composition of two matrix transformations is itself a matrix transformation, and that its standard matrix is the product of the two standard matrices (which also explains why matrix multiplication is defined as it is).

Theorem 1.9.1

If $T_A : R^n \to R^k$ and $T_B : R^k \to R^m$ are matrix transformations, then $T_B \circ T_A$ is also a matrix transformation and

$$T_B \circ T_A = T_{BA} \tag{2}$$

Proof. We first show $T_B \circ T_A$ is linear. For vectors $\mathbf{x}, \mathbf{y} \in R^n$ and scalar k:

$$(T_B \circ T_A)(\mathbf{x}+\mathbf{y}) = T_B(T_A(\mathbf{x}+\mathbf{y})) = T_B(T_A(\mathbf{x})+T_A(\mathbf{y})) = T_B(T_A(\mathbf{x}))+T_B(T_A(\mathbf{y})) = (T_B \circ T_A)(\mathbf{x})+(T_B \circ T_A)(\mathbf{y})$$

$$(T_B \circ T_A)(k\mathbf{x}) = T_B(T_A(k\mathbf{x})) = T_B(kT_A(\mathbf{x})) = kT_B(T_A(\mathbf{x})) = k(T_B \circ T_A)(\mathbf{x})$$

So $T_B \circ T_A$ is a matrix transformation; call its standard matrix C. Then

$$T_C(\mathbf{x}) = (T_B \circ T_A)(\mathbf{x}) = T_B(T_A(\mathbf{x})) = T_B(A\mathbf{x}) = B(A\mathbf{x}) = (BA)\mathbf{x} = T_{BA}(\mathbf{x})$$

By Theorem 1.8.4, $C = BA$. ■

EXAMPLE 1 | The Standard Matrix for a Composition

Let $T_1 : R^3 \to R^2$ and $T_2 : R^2 \to R^3$ be the linear transformations

$$T_1(x, y, z) = (x + 2y,\, x + 2z - y) \quad \text{and} \quad T_2(x, y) = (3x + y,\, x,\, x - 2y)$$

Find the standard matrices for $T_2 \circ T_1$ and $T_1 \circ T_2$.

SOLUTION: The standard basis vectors for R^3 are $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$, $\mathbf{e}_3 = (0, 0, 1)$, giving

$$T_1(\mathbf{e}_1) = (1, 1), \quad T_1(\mathbf{e}_2) = (2, -1), \quad T_1(\mathbf{e}_3) = (0, 2) \implies A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix}$$

The standard basis vectors for R^2 are $\mathbf{e}_1 = (1, 0)$, $\mathbf{e}_2 = (0, 1)$, giving

$$T_2(\mathbf{e}_1) = (3, 1, 1), \quad T_2(\mathbf{e}_2) = (1, 0, -2) \implies B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 1 & -2 \end{bmatrix}$$

By Theorem 1.9.1, the standard matrix for $T_2 \circ T_1$ is

$$BA = \begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 2 \\ 1 & 2 & 0 \\ -1 & 4 & -4 \end{bmatrix}$$

and the standard matrix for $T_1 \circ T_2$ is

$$AB = \begin{bmatrix} 1 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix}$$

Commutativity of Matrix Transformations

Since $AB = BA$ is not generally true, it is also not generally true that $T_{AB} = T_{BA}$, so in general

$$T_A \circ T_B \neq T_B \circ T_A$$

In those special cases where equality holds, we say T_A and T_B **commute**.

EXAMPLE 2 | Composition Is Not Commutative

Let $T_A : R^2 \to R^2$ be the reflection about the line $y = x$, and $T_B : R^2 \to R^2$ be the orthogonal projection onto the y-axis. From the tables in Section 1.8, the standard matrices are

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad BA = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

Since $AB \neq BA$, $T_A \circ T_B \neq T_B \circ T_A$.

EXAMPLE 3 | Composition of Rotations Is Commutative

Rotating a vector through angle θ_1 and then through θ_2 gives the same result as the reverse order, since in both cases the net rotation angle is $\theta_1 + \theta_2$. To verify algebraically, with rotation matrices $A_1 = R_{\theta_1}$ and $A_2 = R_{\theta_2}$:

$$A_1 A_2 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{bmatrix} \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 \\ \sin\theta_2 & \cos\theta_2 \end{bmatrix} = \begin{bmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) \end{bmatrix} = A_2 A_1$$

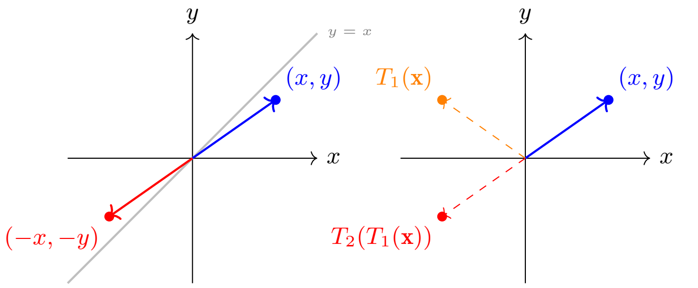
using the addition formulas for sine and cosine. In rotation-matrix notation: $R_{\theta_1} R_{\theta_2} = R_{\theta_1+\theta_2} = R_{\theta_2} R_{\theta_1}$.

EXAMPLE 4 | Composition of Two Reflections

Let $T_1 : R^2 \to R^2$ be the reflection about the y-axis and $T_2 : R^2 \to R^2$ be the reflection about the x-axis. Their standard matrices are $A_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ and $A_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Both $T_1 \circ T_2$ and $T_2 \circ T_1$ have the same effect on any (x, y):

$$(T_1 \circ T_2)(x, y) = T_1(x, -y) = (-x, -y), \qquad (T_2 \circ T_1)(x, y) = T_2(-x, y) = (-x, -y)$$

Both compositions map $\mathbf{x}$ to $-\mathbf{x}$; this is called the **reflection about the origin**.



$T_1 \circ T_2$: reflection about origin

Step-by-step: y-axis then x-axis

For compositions of three or more transformations $T_A : R^n \to R^k$, $T_B : R^k \to R^l$, $T_C : R^l \to R^m$, the standard matrix for $T_C \circ T_B \circ T_A$ is CBA:

$$T_C \circ T_B \circ T_A = T_{CBA} \tag{4}$$

EXAMPLE 5 | Composition of Three Matrix Transformations

Find the image of $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ under the composition: first rotate by $\pi/6$, then reflect about $y = x$, then project orthogonally onto the y-axis.

SOLUTION: From the tables in Section 1.8, the standard matrices are

$$A = \begin{bmatrix} \cos(\pi/6) & -\sin(\pi/6) \\ \sin(\pi/6) & \cos(\pi/6) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

The standard matrix for $T_C \circ T_B \circ T_A$ is

$$CBA = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$$

Thus the image is $\begin{bmatrix} 0 \\ \frac{\sqrt{3}}{2}x - \frac{1}{2}y \end{bmatrix}$.

Invertibility of Matrix Operators

If $T_A : R^n \to R^n$ is a matrix operator whose standard matrix A is invertible, then T_A is called **invertible**, and its **inverse** is

$$T_A^{-1} = T_{A^{-1}} \tag{5}$$

This is the transformation whose standard matrix is A^{-1}. From (5) and Theorem 1.9.1:

$$T_A^{-1} \circ T_A = T_{A^{-1}A} = T_I \quad \text{and} \quad T_A \circ T_A^{-1} = T_{AA^{-1}} = T_I$$

so T_A^{-1} undoes the effect of T_A.

EXAMPLE 6 | Inverse of a Rotation Operator

Let $T : R^2 \to R^2$ be the counterclockwise rotation by angle θ, with standard matrix $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

SOLUTION: Geometrically, undoing a counterclockwise rotation by θ requires a clockwise rotation by θ (i.e., a rotation by $-\theta$). Indeed, from (5) and Theorem 1.4.5:

$$R_\theta^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} = R_{-\theta}$$

EXAMPLE 7 | Inverse Transformations from Linear Equations

Consider the operator $T : R^2 \to R^2$ defined by $w_1 = 2x_1 + x_2$, $w_2 = 3x_1 + 4x_2$. Find $T^{-1}(w_1, w_2)$.

SOLUTION: The standard matrix is $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, which is invertible, so

$$A^{-1} = \begin{bmatrix} \frac{4}{5} & -\frac{1}{5} \\ -\frac{3}{5} & \frac{2}{5} \end{bmatrix}$$

Thus the inverse transformation is

$$T^{-1}\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = A^{-1}\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} \frac{4}{5}w_1 - \frac{1}{5}w_2 \\ -\frac{3}{5}w_1 + \frac{2}{5}w_2 \end{bmatrix}$$

i.e., $T^{-1}(w_1, w_2) = \left(\frac{4}{5}w_1 - \frac{1}{5}w_2,\ -\frac{3}{5}w_1 + \frac{2}{5}w_2\right)$.

Remark. Not every matrix has an inverse; correspondingly, not every matrix transformation has an inverse. For example, the orthogonal projection operators in Section 1.8 have non-invertible standard matrices and thus have no inverse transformation.

Exercise Set 1.9

1. Determine whether the operators T_1 and T_2 commute (i.e., whether $T_1 \circ T_2 = T_2 \circ T_1$).
   (a) $T_1 : R^2 \to R^2$ is the reflection about the line $y = x$, and $T_2 : R^2 \to R^2$ is the orthogonal projection onto the x-axis.
   (b) $T_1 : R^2 \to R^2$ is the reflection about the x-axis, and $T_2 : R^2 \to R^2$ is the reflection about the line $y = x$.

2. Determine whether the operators commute.
   (a) $T_1 : R^2 \to R^2$ is the orthogonal projection onto the x-axis, and $T_2 : R^2 \to R^2$ is the orthogonal projection onto the y-axis.
   (b) $T_1 : R^2 \to R^2$ is the rotation through angle $\pi/4$, and $T_2 : R^2 \to R^2$ is the reflection about the y-axis.

3. $T_1 : R^3 \to R^3$ is the reflection about the xy-plane, and $T_2 : R^3 \to R^3$ is the orthogonal projection onto the yz-plane. Determine whether T_1 and T_2 commute.

4. $T_1 : R^3 \to R^3$ is the reflection about the xy-plane, and $T_2 : R^3 \to R^3$ is defined by $T_2(x, y, z) = (2x, 3y, z)$. Determine whether T_1 and T_2 commute.

5. Let T_A and T_B be operators whose standard matrices are given. Find the standard matrices for $T_B \circ T_A$ and $T_A \circ T_B$.
$$A = \begin{bmatrix} 1 & -2 \\ 4 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & -3 \\ 5 & 0 \end{bmatrix}$$

7. Find the standard matrix for the stated composition in R^2.
   (a) A rotation of 90°, followed by a reflection about the line $y = x$.
   (b) An orthogonal projection onto the y-axis, followed by a 45° counterclockwise rotation.
   (c) A reflection about the x-axis, followed by a rotation of 60°.

8. Find the standard matrix for the stated composition in R^2.
   (a) A rotation of 60°, followed by an orthogonal projection onto the x-axis, followed by a reflection about the line $y = x$.
   (b) A rotation of 15°, followed by a rotation of 105°, followed by a rotation of 60°.

9. Find the standard matrix for the stated composition in R^3.
   (a) A reflection about the yz-plane, followed by an orthogonal projection onto the xz-plane.
   (b) A reflection about the xy-plane, followed by an orthogonal projection onto the xy-plane.

11. Let $T_1(x_1, x_2) = (x_1 + x_2,\ x_1 - x_2)$ and $T_2(x_1, x_2) = (3x_1,\ 2x_1 + 4x_2)$.
   (a) Find the standard matrices for T_1 and T_2.
   (b) Find the standard matrices for $T_2 \circ T_1$ and $T_1 \circ T_2$.
   (c) Use the matrices from (b) to find formulas for $T_1(T_2(x_1, x_2))$ and $T_2(T_1(x_1, x_2))$.

13. Let $T_1(x_1, x_2) = (x_1 - x_2,\ 2x_2 - x_1,\ 3x_1)$ and $T_2(x_1, x_2, x_3) = (4x_2,\ x_1 + 2x_2)$.
   (a) Find the standard matrices for T_1 and T_2.
   (b) Find the standard matrix for $T_2 \circ T_1$ and $T_1 \circ T_2$.
   (c) Use the matrices from (b) to find formulas for $T_1(T_2(x_1, x_2, x_3))$ and $T_2(T_1(x_1, x_2))$.

17. Express the equations in matrix form; then use Theorem 1.5.3(c) to determine whether the operator defined by the equations is invertible.

(a) $w_1 = 8x_1 + 4x_2,\ w_2 = 2x_1 + x_2$

(b) $w_1 = -x_1 + 3x_2 + 2x_3,\ w_2 = 2x_1 + 4x_3,\ w_3 = x_1 + 3x_2 + 6x_3$

19. Determine whether the operator $T : R^2 \to R^2$ is invertible; if so, find the standard matrix for T^{-1} and find $T^{-1}(w_1, w_2)$.

(a) $w_1 = x_1 + 2x_2,\ w_2 = -x_1 + x_2$

(b) $w_1 = 4x_1 - 6x_2,\ w_2 = -2x_1 + 3x_2$

21. Determine whether the matrix operator is invertible. If so, describe the effect of its inverse in words.

(a) Reflection about the x-axis in R^2.

(b) A 60° counterclockwise rotation about the origin in R^2.

(c) Orthogonal projection onto the x-axis in R^2.

23. Determine whether the operator T_A is invertible. If so, compute $T_A^{-1}(\mathbf{x})$.

(a) $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix};\ \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix};\ \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

25. Let $T_A : R^2 \to R^2$ be multiplication by

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

(a) Describe the geometric effect of applying T_A to a vector $\mathbf{x}$ in R^2.

(b) Express the operator T_A as a composition of two linear operators on R^2.

27. Prove that the matrix transformations T_A and T_B commute if and only if the matrices A and B commute.

28. Let T_A and T_B be matrix operators on R^n. Prove that $T_A \circ T_B$ is invertible if and only if both T_A and T_B are invertible.

29. Prove that the matrix operator T_A on R^n is invertible if and only if for every vector $\mathbf{b}$ in R^n there exists a unique vector $\mathbf{x}$ in R^n such that $T_A(\mathbf{x}) = \mathbf{b}$.